Example 3.3 Eliminate θ from $a\cos\theta = b$ and $c\sin\theta = d$, where a, b, c, d are constants.

Solution:
Squaring and adding $ac\cos\theta = bc$ and $ac\sin\theta = ad$, we get

$$a^2c^2 = b^2c^2 + a^2d^2.$$



Exercise - 3.1

1. Identify the quadrant in which an angle of each given measure lies

   (i) 25° (ii) 825° (iii) -55° (iv) 328° (v) -230°

2. For each given angle, find a coterminal angle with measure of θ such that $0^\circ \leq \theta < 360^\circ$

   (i) 395° (ii) 525° (iii) 1150° (iv) -270° (v) -450°

3. If $a\cos\theta - b\sin\theta = c$, show that $a\sin\theta + b\cos\theta = \pm\sqrt{a^2 + b^2 - c^2}$.

4. If $\sin\theta + \cos\theta = m$, show that $\cos^6\theta + \sin^6\theta = \dfrac{4 - 3\left(m^2 - 1\right)^2}{4}$, where $m^2 \leq 2$.

5. If $\dfrac{\cos^4\alpha}{\cos^2\beta} + \dfrac{\sin^4\alpha}{\sin^2\beta} = 1$, prove that

   (i) $\sin^4\alpha + \sin^4\beta = 2\sin^2\alpha\sin^2\beta$ (ii) $\dfrac{\cos^4\beta}{\cos^2\alpha} + \dfrac{\sin^4\beta}{\sin^2\alpha} = 1.$

6. If $y = \dfrac{2\sin\alpha}{1 + \cos\alpha + \sin\alpha}$, then prove that $\dfrac{1 - \cos\alpha + \sin\alpha}{1 + \sin\alpha} = y.$

7. If $x = \sum_{n=0}^{\infty}\cos^{2n}\theta$, $y = \sum_{n=0}^{\infty}\sin^{2n}\theta$ and $z = \sum_{n=0}^{\infty}\cos^{2n}\theta\sin^{2n}\theta$, $0 < \theta < \dfrac{\pi}{2}$, then show that $xyz = x + y + z$.
   [Hint: Use the formula $1 + x + x^2 + x^3 + \ldots = \dfrac{1}{1 - x}$, where $|x| < 1$].

8. If $\tan^2\theta = 1 - k^2$, show that $\sec\theta + \tan^3\theta\ \text{cosec}\ \theta = \left(2 - k^2\right)^{3/2}$. Also, find the values of k for which this result holds.

9. If $\sec\theta + \tan\theta = p$, obtain the values of $\sec\theta, \tan\theta$ and $\sin\theta$ in terms of p.

10. If $\cot\theta\left(1 + \sin\theta\right) = 4m$ and $\cot\theta\left(1 - \sin\theta\right) = 4n$, then prove that $\left(m^2 - n^2\right)^2 = mn$.

11. If $\text{cosec}\ \theta - \sin\theta = a^3$ and $\sec\theta - \cos\theta = b^3$, then prove that $a^2b^2\left(a^2 + b^2\right) = 1$.

12. Eliminate θ from the equations $a\sec\theta - c\tan\theta = b$ and $b\sec\theta + d\tan\theta = c$.

3.3 Radian Measure



Initially right triangles were used to define trigonometric ratios and angles were measured in degrees. But right triangles have limitations as they involve only acute angles. In degrees a full rotation corresponds to 360° where the choice of 360 was made thousands of years prior to the Babylonians. They might have chosen 360 based on the number of days in a year. But it does have the nice property of breaking into smaller angles like $30^\circ, 45^\circ, 60^\circ, 90^\circ$ and 180°. In 17th century,
trigonometry was extended to Physics and Chemistry where it required trigonometric functions whose domains were sets of real numbers rather than angles. This was accomplished by using correspondence between an angle and length of the arc on a unit circle. Such a measure of angle is termed as **radian measure** . For theoretical applications, the radian is the most common system of angle measurement. Radians are common unit of measurement in many technical fields, including calculus. The most important irrational number π plays a vital role in radian measures of angles. Let us introduce the radian measure of an angle.